

# Zhong Wang

Ph.D. | Computational Biologist & Genome AI Research Lead

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✉️ [zhongwang@lbl.gov](mailto:zhongwang@lbl.gov) |  [Google Scholar](#) |  Ph.D., Duke University (2004)

## PROFILE

Computational biologist and research group lead specializing in **genome foundation models**, large-scale metagenomics, and scalable machine learning for the life sciences. Creator of **GenomeOcean**, an efficient Mixture-of-Experts genome foundation model, and architect of open-source genomic AI tooling adopted across the research community. Twenty-plus years bridging high-performance computing, deep learning, and microbial genomics, with a sustained record of mentorship, multi-institutional collaboration, and competitive resource acquisition.

## PROFESSIONAL EXPERIENCE

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| 2009–<br>present | <b>Career Computational Biologist &amp; Genome Analysis Group Lead</b> , DOE Joint Genome Institute; Environmental Genomics & Systems Biology Division, Lawrence Berkeley National Laboratory, Berkeley, CA.                               |
| 2022–<br>present | <b>Adjunct Professor</b> , Dept. of Molecular & Cellular Biology, School of Natural Sciences; Founder & Director, JGI/UC Merced Distinguished Summer Internship Program, University of California, Merced, CA.                             |
| 2023–2026        | <b>Google Cloud Innovator &amp; Google Developer Expert</b> .                                                                                                                                                                              |
| 2022–2025        | <b>Senior Advisor</b> , MemVerge Inc. (Cloud Automation Platform Service), Milpitas, CA.                                                                                                                                                   |
| 2013–2022        | <b>Adjunct Associate Professor</b> & Founder/Director, JGI/UC Merced Summer Internship Program, University of California, Merced, CA.                                                                                                      |
| 2008–2009        | <b>Associate Research Scientist</b> (Dr. Michael Snyder), Dept. of Molecular, Cellular & Developmental Biology, Yale University; <b>Director of Bioinformatics</b> , Yale Stem Cell Center, Yale University Medical School, New Haven, CT. |
| 2004–2007        | <b>Postdoctoral Fellow</b> (Dr. Huntington F. Willard), Institute for Genome Science & Policy, Duke University, Durham, NC.                                                                                                                |
| 1997–2004        | <b>Research Assistant</b> (Dr. Haifan Lin), Dept. of Cell Biology, Duke University, Durham, NC.                                                                                                                                            |

## EDUCATION

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| 2004 | <b>Ph.D., Cell Biology</b> — Duke University, Durham, NC.                                                              |
| 2002 | <b>Certificate, Bioinformatics &amp; Genome Technology</b> — Duke University, Durham, NC.                              |
| 1997 | <b>M.S., Developmental Biology</b> — Shanghai Institute of Cell Biology, Chinese Academy of Sciences, Shanghai, China. |
| 1994 | <b>B.S., Microbiology</b> — Shandong University, Ji'nan, China.                                                        |

## Executive & Professional Development

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| 2023 | Certified Innovator Program on the Google Cloud Platform, Google Inc.                       |
| 2021 | Transformative Leadership Development Program, Berkeley Lab Stewardship Summit.             |
| 2012 | Leadership Development Program, Center for Executive Education, UC Berkeley & Berkeley Lab. |

## HONORS & FELLOWSHIPS

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| 2021      | Director's Award, Lawrence Berkeley National Laboratory.                                                      |
| 2008      | "Crack the Code with the Bear" Pediatric Cancer Research Award.                                               |
| 1997–2004 | James B. Duke Fellowship.                                                                                     |
| 1990–1994 | Outstanding Undergraduate Student Fellowship; First place, college entrance exam, Shandong University (1990). |

## SELECTED RESEARCH HIGHLIGHTS

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- **GenomeOcean.** Designed a microbial genome foundation model using an 8-expert Mixture-of-Experts (MoE) architecture; sparse-upcycling a dense model delivered a **1.87× inference throughput gain** and higher biological plausibility while using **2.5× fewer active parameters**. (*Major revision in progress, Nature Methods.*)
- **Routing-aware probing.** Showed that MoE routing vectors capture functional biological specialization (e.g. tRNA / rRNA classification) without explicit supervision — a new tool for unsupervised functional annotation.
- **Genome-Factory.** Built and maintains an open-source library unifying genomic foundation-model workflows (data prep, model loading, benchmarking); **accepted to ICML 2026**, 130+ GitHub stars.
- **genomeocean-sentinel (GOS) v1.0.** AI-detection framework for synthetic DNA signatures; LBNL software disclosure #2026-126 (under review for open-source release).

## PUBLICATIONS

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Full list: [Google Scholar](#). (\*co-first author)

### Books & Book Chapters

1. Wang Z, Sannyasi A, Jiang J (Eds.). **Transforming Life Sciences with Cloud Computing and AI**. Springer Nature, 2026. Print ISBN 978-3-032-14221-4; eBook ISBN 978-3-032-14222-1. DOI: 10.1007/978-3-032-14222-1. (*Published Apr 2026.*)
2. Wang Z (Ed.). **Introduction to Computational Metagenomics**. 2022. ISBN 978-981-124-246-5. DOI: 10.1142/12425.
3. Johnson NV, Wang Z. *Transcriptomics: Next Generation Transcriptome*. In **Encyclopedia of Drug Metabolism and Interactions** (Ed. Lyubimov AV), 2017. DOI: 10.1002/9780470921920.edm149.

### Selected Journal Articles & Proceedings

1. Zhou Z, Wu W, Ho H, Wang J, Shi L, Davuluri RV, **Wang Z**, Liu H. DNABERT-S: Learning species-aware DNA embedding with genome foundation models. **Bioinformatics**, 2025, 41(7).

DOI: 10.1093/bioinformatics/btaf188.

2. Wu W, Song X, Wen Y, Lin Q, Zhou Z, Huo J, **Wang Z**, Liu H. Genome-Factory: A Library for Tuning, Deploying, and Interpreting Genomic Foundation Models. **Proc. 43rd Int. Conf. on Machine Learning (ICML)**, 2026. (*Accepted.*)
3. Ho H, Chovatia M, Egan R, He G, Yoshinaga Y, Liachko I, O'Malley R, **Wang Z**. Integrating chromatin conformation information in a self-supervised learning model improves metagenome binning. **PeerJ**, 2023, 11:e16129.
4. Sun J, Egan RE, Ho H, Li Y, **Wang Z**. Persistent Memory as an Effective Alternative to RAM in Metagenome Assembly. **BMC Bioinformatics**, 2022, 23(1).
5. Meyer F, Fritz A, Deng ZL, Koslicki D, et al. Critical Assessment of Metagenome Interpretation — second round of challenges. **Nature Methods**, 2022, 19(4):429–440.
6. Li K, Lu Y, Deng L, Wang L, Shi L, **Wang Z**. Deconvolute individual genomes from metagenome sequences through short-read clustering. **PeerJ**, 2020, 8:e8966.
7. Shi L, **Wang Z**. Computational Strategies for Scalable Genomics Analysis. **Genes**, 2019, 10(12):1017.
8. Lapierre N, Egan R, Wang W, **Wang Z**. MiniScrub: de novo long-read scrubbing using approximate alignment and deep learning. **BMC Bioinformatics**, 2019, 20(1):552.
9. Shi L, Meng X, Tseng E, Mascagni M, **Wang Z**. SpaRC: Scalable Sequence Clustering using Apache Spark. **Bioinformatics**, 2018. DOI: 10.1093/bioinformatics/bty733.
10. Sczyrba A, Hofmann P, Belmann P, et al. Critical Assessment of Metagenome Interpretation: a benchmark of metagenomics software. **Nature Methods**, 2017, 14(11):1063–1071.
11. Kang DD, Froula J, Egan R, **Wang Z**. MetaBAT: efficiently reconstructing single genomes from complex microbial communities. **PeerJ**, 2015, 3:e1165.
12. Gordon SP, Tseng E, Salamov A, et al, **Wang Z**. Widespread Polycistronic Transcripts in Fungi Revealed by Single-Molecule mRNA Sequencing. **PLoS ONE**, 2015, 10(7):e0132628.
13. Shi W, Moon C, Leahy S, Kang D, et al, **Wang Z**, Attwood G, Rubin E. Methane yield phenotypes linked to differential gene expression in the sheep rumen microbiome. **Genome Research**, 2014.
14. Martin JA, Johnson NV, Gross SM, et al, **Wang Z**. A near-complete snapshot of the *Zea mays* seedling transcriptome from ultra-deep sequencing. **Scientific Reports**, 2014. DOI: 10.1038/srep04519.
15. Byrne RT, Klingele AJ, Cabot EL, et al, **Wang Z**, et al. Evolution of Extreme Resistance to Ionizing Radiation via Genetic Adaptation of DNA Repair. **eLife**, 2014.
16. Nordberg H, Bhatia K, Wang K, **Wang Z**. BioPig: a Hadoop-based analytic toolkit for large-scale sequence data. **Bioinformatics**, 2013. DOI: 10.1093/bioinformatics/btt528.
17. Clark S, Egan R, Frazier PI, **Wang Z**. ALE: a Generic Assembly Likelihood Evaluation Framework. **Bioinformatics**, 2013. DOI: 10.1093/bioinformatics/bts723.
18. Koren S, Schatz MC, Walenz BP, et al, **Wang Z**, et al, Phillippy AM. Hybrid error correction and de novo assembly of single-molecule sequencing reads. **Nature Biotechnology**, 2012.
19. Martin J, **Wang Z**. Next-generation transcriptome assembly. **Nature Reviews Genetics**, 2011, 12(10):671–682.
20. Hess M, Sczyrba A, Egan R, et al, **Wang Z**, Rubin EM. Metagenomic discovery of biomass-degrading genes and genomes from cow rumen. **Science**, 2011, 331(6016):463–467.
21. Bruno VM, **Wang Z**, Marjani SL, et al, Snyder M. Comprehensive annotation of the *Candida albicans* transcriptome using RNA-seq. **Genome Research**, 2010, 20(10):1451–1458.
22. He S, Wurtzel O, Singh K, et al, **Wang Z**, et al, Hugenholtz P. Validation of two rRNA removal methods for microbial metatranscriptomics. **Nature Methods**, 2010, 7(10):807–812.

23. Raha D, **Wang Z**, Moqtaderi Z, et al, Snyder M. Close association of RNA polymerase II and many transcription factors with Pol III genes. **PNAS**, 2010, 107(8):3639–3644.
24. **Wang Z**, Gerstein M, Snyder M. RNA-Seq: a revolutionary tool for transcriptomics. **Nature Reviews Genetics**, 2009, 10(1):57–63.
25. Nagalakshmi U, **Wang Z\***, Waern K, et al, Snyder M. The transcriptional landscape of the yeast genome defined by RNA sequencing. **Science**, 2008, 320(5881):1344–1349.
26. Grivna ST, Beyret E, **Wang Z**, Lin H. A novel class of small RNAs in mouse spermatogenic cells. **Genes & Development**, 2006, 20(13):1709–1714.
27. **Wang Z**, Lin H. Nanos maintains germline stem cell self-renewal by preventing differentiation. **Science**, 2004, 303(5666):2016–2019.

*Additional peer-reviewed articles (selected from a record of 46): Kang D, Froula J, Egan R, Wang Z. MetaBAT (PeerJ, 2019); Lin H, Su Z, Meng X, et al, Wang Z (Int. J. Parallel Programming, 2018); Shi L, Wang Z, Yu W, Meng X (Parallel Computing, 2017); Kang DD, Rubin EM, Wang Z (it-Information Technology, 2016, invited); Wang Z, Willard HF (BMC Genomics, 2012); Fang Z, Martin JA, Wang Z (Cell & Bioscience, 2012); Wang Z, Willard HF, Mukherjee S, Furey TS (PLoS Comp. Biol., 2006); Wang Z, Lin H (Current Biology, 2005); Szakmary A, Cox DN, Wang Z, Lin H (Current Biology, 2005); Parisi MJ, Deng W, Wang Z, Lin H (Genesis, 2001); and others. Full list on Google Scholar.*

### Preprints & Manuscripts in Review

1. Wu W, Zhou Z, Riley R, et al, Shi L, Liu H, **Wang Z**. Uncovering the Genomic Manifold via Scalable Learning from the Global Microbiome (GenomeOcean). **bioRxiv**, 2025.01.30.635558. DOI: 10.1101/2025.01.30.635558. (*Nature Methods*, revision in progress.)
2. **Wang Z**, Yang G, Liu F, Egan R. Positional Attention Signatures Distinguish AI-Generated from Natural Biological Sequences. (*Submitted, NeurIPS 2026.*)
3. Yang G, Ghasemian A, Liu F, **Wang Z**, Mehrabi N, Kabra S, Hosseinmardi H. Asking Back: Interaction-Layer Antidistillation Watermarks. (*Submitted, NeurIPS 2026.*)
4. Yang G, Liu F, Ghasemian A, **Wang Z**, Mehrabi N, Hosseinmardi H. Asymmetric Phase Coding Audio Watermarking. (*Submitted, NeurIPS 2026.*)
5. Wu W, Song X, Wen Y, Lin Q, Zhou Z, Hu JYC, **Wang Z**, Liu H. Genome-Facory: An Integrated Library for Tuning, Deploying, and Interpreting Genomic Models. **arXiv**:2509.12266, 2025.
6. Lettich R, Egan R, Riley R, **Wang Z**, Tritt A, Olikier L, Yelick K, Buluc A. GenomeFace: a deep-learning metagenome binner trained on 43,000 microbial genomes. **bioRxiv**, 2024.02.07.579326.

### SELECTED INVITED TALKS & SEMINARS

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| <b>2026.04</b> | <b>Invited Seminar</b> , Tulane University School of Medicine (Biochemistry & Molecular Biology). <i>Genome Foundation Models and the Genomic Manifold Hypothesis</i> . New Orleans, LA.                      |
| <b>2025.11</b> | <b>Invited Speaker</b> , Joint Berkeley Initiative for Microbiome Sciences (JBIMS) Workshop #2. <i>Advanced Topics Using Foundation Genome Models: Inference, Evaluation, and Fine-tuning</i> . Berkeley, CA. |
| <b>2025.10</b> | <b>Invited Speaker</b> , JBIMS Workshop #1 (AI in Microbiome Science). <i>GenomeOcean: An Efficient Genome Foundation Model</i> . Berkeley, CA.                                                               |
| <b>2025.05</b> | <b>SFA<sup>2</sup>F</b> (Sequence to Function). <i>GenomeOcean: An Efficient Genome Foundation Model</i> . Santa Fe, NM.                                                                                      |
| <b>2024.11</b> | LA CaTS Center, Biostatistics & Epidemiology Seminar. <i>GenomeOcean, A Pre-trained Metagenomic Foundational Model</i> . Virtual.                                                                             |

- 2024.02** Northwestern University, Dept. of Statistics & Data Science. *From 20GB to 100TB: a journey on the metagenomic road*. Chicago, IL.
- 2023.07** MICROBIOME — ISMB/ECCB 2023. *Axolotl: A Scalable Apache Spark Library for High-throughput Genomic Data Analysis*. Lyon, France.
- 2022.07** ISMB 2022. *Persistent Memory as an Alternative to RAM in Metagenome Assembly*. Madison, WI.

30+ additional invited talks (2004–2021) at venues including AGBT, CSHL Genome Informatics, Spark Summit, Stanford, UC Berkeley, and international symposia in China and Germany.

## RESEARCH SUPPORT

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- 2009–present** **co-PI** — DOE BER Field Work Proposal to DOE JGI (DE-AC02-05CH112), ~\$1M/yr.
- 2025–2026** **Co-PI** — DOE Genesis Mission Proposal (OS4LS; Northwestern lead), ~\$150k. (*Submitted Apr 2026.*)
- 2025–2026** **Co-PI** — Strategic Partnership Project (SPP), UC Merced, ~\$30k. (*Ongoing.*)
- 2024–2025** **PI** — NERSC DDR “GenAI for Science”: Biosynthetic Gene Cluster Foundation Model (bgcFM), 30k GPU node-hours.
- 2024** **PI** — NERSC ERCAP/DDR: Impact of Tokenization Strategy on Genome Large Models, 10k GPU node-hours.
- 2021** **PI** — Google Inc., Spark-based metagenome clustering pilot on GCP, \$20k.
- 2017–2018** **PI** — NSF XSEDE Startup (MCB170069), Pittsburgh Supercomputing Center, ~\$15k.
- 2015–2016** **PI** — Amazon Web Services Education Research Grant, \$20k.
- 2009–2011** **PI** — DOE BER Field Work Proposal: Cow Rumen Metagenome Informatics, \$1M.
- 2011–2013** **PI** — LBNL LDRD: Machine learning + HPC to discover novel enzymes, \$334k.
- 2010–2012** **co-PI** — LBNL LDRD: Transcriptome Analysis of Agave, a candidate biofuel feedstock.

## PROFESSIONAL SERVICE & KNOWLEDGE TRANSFER

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- **Abstract Chair**, ISMB 2026 Microbiome COSI (Washington, D.C.) — led abstract selection, reviewer coordination, and program scheduling.
- **Program Committee**, ISMB/ECCB 2018–2026.
- **Peer Reviewer**, NeurIPS 2026 (computational biology) and ACS Synthetic Biology (2026).
- **Graduate Admissions Committee**, UC Merced (2019, 2026).
- Ad-hoc grant review panels for **NIDDK** and **NSF**.
- **Software & Technology Transfer**: Genome-Factory (ICML 2026; 130+ GitHub stars); genomeocean-sentinel v1.0 (LBNL software disclosure #2026-126, under review for open-source release).

## Editorial Responsibilities

- Editor, *Nature Scientific Reports* (2012–2024).

- Editor, Special Issue “Impact of Parallel and High-performance Computing in Genomics,” *Genes* (2019).
- Guest Editor, *BMC Cell & Biosciences* (2012).

## MENTORSHIP & TRAINING

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**Group composition (FY26):** 1 postdoc, 1 research associate/technician, 3 affiliates/guests.

### Postdoctoral Scholars

- **Dongwan (Don) Kang** (2012–2013) — VP for AI, Samsung.
- **Weibing Shi** (2011–2013) — Scientific Director, Computational Biology, Amgen.
- **Alex Sczyrba** (2009–2011) — Director, Computational Metagenomics, University of Bielefeld / Forschungszentrum Jülich, Germany.
- **Changbin Du** (2009–2011) — Biostatistician, Truvian.

### Graduate Students & Thesis Committees

- **Zhihan Zhou** (thesis committee, Northwestern, 2024–2025; *Ph.D. defended FY26*) — Research Scientist, NVIDIA.
- **Harrison Ho** (thesis advisor, UC Merced, 2019–2024) — Postdoc, Albert Einstein College.
- **Lizhen Shi** (external thesis advisor, Florida State, 2017–2020) — Assistant Professor, Northwestern.
- **Chantz Williams** (thesis committee, UC Merced, 2024–present, ongoing).
- Thesis committees: Travis Lawrence, Steve Wilson, Thaddeus Seher, Josephine Sami (UC Merced, 2014–2022).

### Undergraduate & Internship Pipeline

- **FY26 cohort:** Oliver Benelli (UC Berkeley CalTeach), Sally Liao (Northwestern SULI), Brandon Barana (UC Merced).
- Founder & Director, **JGI/UC Merced Distinguished Summer Internship Program**; co-lead, **DOE RENEW** pipeline for underrepresented students in science.
- 40+ mentored interns since 2010 across the DOE Computational Biology Graduate Fellowship, LBNL SULI, CalTeach, and JGI-UCM programs.